



Center of Science and Technology Advanced Development
Academic Year: 2024
DevOps Engineering
Course Syllabus
Duration: 70 hours

COURSE DESCRIPTION

This comprehensive course is designed to equip you with advanced knowledge in cloud application development. You'll gain a deep understanding of best practices for deploying applications on servers, ensuring optimal performance and scalability. Additionally, you'll master essential DevOps tools that streamline the software development lifecycle, automate processes, and enhance the efficiency and delivery of your applications.

COURSE OUTLINE

N°	Content	Hours	Related D's
1	DevOps Terminology - α Story of DevOps - α What and Why DevOps? - α Prerequisite - α Software Development Methodology - α DevOps Methodology - α DevOps Tools and Roadmaps Lab: Theory and Discussion	2	D001
2	Operating Systems Concepts - α What is an Operating System? - α What is Kernel? - α Function of Operating System - α Groups of Operating System Lab: Theory	2	D002

3	Managing Server - α Introduction to Server - α Terminal & Editor - α Linux Introduction - α Linux Basic Commands - α User & Group Management - α SystemD & Crontab - α Shell Script - α Lesson Recap* Lab: Classwork	4	D003
4	Networking, Security and Protocols - α Security Overview - α Understanding HTTP, NFS, SSH, etc. - α Understanding Reverse Proxy, Forward Proxy, Load Balancing, Firewall, etc. - α Setup NFS Lab: Homework	2	D004
5	Web Server - α What is Web Server? - α Web Server Setup & Configuration - α NGINX, Apache - α Deploy Application Lab: Classwork	4	D005
6	Containerized - α What is Containerization? - α Docker Introduction - α Docker Images, Dockerfile, Docker Containers - α Deploy Web Application using Docker - α Domain Name Configuration Lab: Homework	6	D006
7	Docker Ecosystem - α What is Docker Storage? - α Type of Mounting Data in Docker - α Database Server and Data Backup - α What and Why Compose? - α Compose Installation - α YAML file - α Deploy Microservices app using Compose Lab: Presentation	6	D007

8	Nexus Repository Manager - What is NXRM? - Nexus Installation - User and Role - Blob Store - Setup Docker Repo - Setup Helm Repo	4	D008
9	CI / CD - Introduction to CI / CD - CI / CD Tools and What are they? - What's the difference between CI and CD (and the other CD)? - Jenkins Introduction - Jenkins Architecture - Jenkins Workspace and Jobs Lab: Practical (Auto Deployment)	8	D009
10	CI / CD Pipeline - Jenkins Shared Library - Managing Nodes on Jenkins - Understanding CI/CD pipeline - Some configuration in Jenkins - Creating an end-to-end automated CI/CD Pipeline	4	D010
11	Assignment - Assign topic for student to research, prepare slide for presentation Topic Cover: - Automate deployment Lab: Presentation	4	D013
12	Configuration Management - Overview - How does Ansible work? - Ansible Architecture - Ansible Playbooks - Ansible Roles - Applying configuration using Ansible Lab: Practical	4	D014

13	Container Orchestration I - α Introduction to Kubernetes - α Features of Kubernetes - α Why Kubernetes? - α Docker Swarm vs Kubernetes - α What is Pod in Kubernetes? - α Kubernetes Configuration - α Creating a Deployment - α Creating a Service - α Type of Services - α Creating an Ingress - α Demonstrating the use of Ingress, services and deployments together - α Installing Kubernetes Dashboard Lab: Classwork	4	D015
14	Container Orchestration II - α Deploying Kubernetes using Kubeadms - ο Configmaps - ο Secret - α Understanding storage in Kubernetes - ο hostPath - ο NFS - α Kubernetes + Jenkins Lab: Classwork	4	D016
15	Argo CD and Helm Chart - α What is ArgoCD? - α How does it work? - α Argo Webhooks - α Kubernetes + Argo - α What is a Helm Chart? - α Helm Command - α Helm Chart Usage Lab: Classwork	4	D017
16	Continuous Monitoring - α What is a Monitoring tool? - α Prometheus Node with exporter - α Visualize node with Grafana - α Uptime Kuma Lab: Classwork	4	D018
17	Cloud Provider Services - α Cloud Providers Overview - α Introduction to AWS & GCP - α AWS & GCP Services Usage Lab: Mini Project	2	D019

18	Deployment & Final Project: a Apply What we learn a Assign Final Project {Topic will be provided}	2	D022
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TEACHING AND LEARNING ACTIVITIES

1. Attendance in class and completion of all tasks, quizzes and assignments is required and counts towards your final score.
2. If you miss a class, you must catch up on all the content and ensure that you understand the content. It is your obligation to ask the lecturer or classmates about any class you missed.
3. You are expected to pay attention to what is being taught during the class. Please be respectful when your classmates, or lecturer are presenting.

ASSESSMENT METHOD

Coursework will be assessed as follows:

Attendance & Class Participant:	10%
Classwork & Homework:	20%
Assignment & Presentation:	20%
Mini Project / Midterm Exam:	20%
Final Project / Exam:	30%

Note: Attendance requirement!

Students should attend more than **60%** of the course.

In case the attendance is lower than **60%**, and if the student fails the exam, the student will not be able to get a certificate from our institute.

Grading Scale		
Score	Grade	Explanation
90-100	A	Excellent
85-89	B+	Good
80-84	B	Above Average
75-79	C+	Average
70-74	C	Below Average

60-69	D	Poor
≤ 59	F	Fail

DevOps Tools





Jenkins



ANSIBLE



kubernetes



argo



Grafana



Uptime Kuma



Prometheus

